

1.3.2 Examples

Ex 1: let be $F: \mathbb{R}^3 \rightarrow \mathbb{R}^3$
 $(x, y, z) \mapsto F(x, y, z) = (x^2 - e^z, \sin(xz), y^2 e^{2xz})$

$$\operatorname{div} F(x, y, z) = 2x + 0 + 2xy^2 e^{2xz} = 2x(1 + y^2 e^{2xz}) \in \mathbb{R}$$

Ex 2: let us consider the gravitational force (vector field)

$$F: \mathbb{R}^3 \setminus (0, 0, 0) \rightarrow \mathbb{R}^3$$

$$(x, y, z) \mapsto F(x, y, z) = -\frac{c}{r^3} (x, y, z)$$

Note:

$$\begin{aligned} P_0 &= (x_0, y_0, z_0) \\ &= (0, 0, 0) \end{aligned}$$

$$\operatorname{div} F(x, y, z) = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}$$

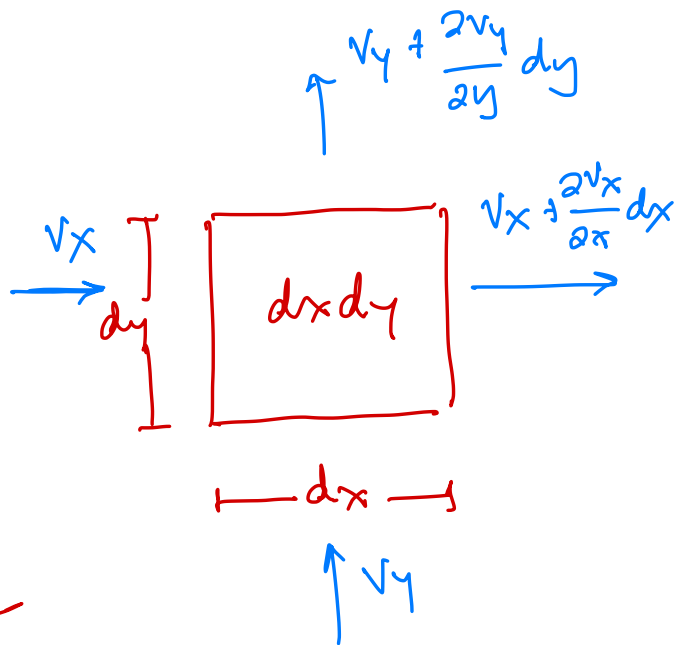
$$\frac{\partial F_1}{\partial x} = \frac{\partial}{\partial x} \left(-\frac{c}{r^3} x \right) = c \frac{3x^2 - r^2}{r^5}$$

$$\begin{aligned} \operatorname{div} F(x, y, z) &= \frac{c}{r^5} (3x^2 - r^2 + 3y^2 - r^2 + 3z^2 - r^2) \\ &= \frac{c}{r^5} (3(\underbrace{x^2 + y^2 + z^2}_{r^2}) - 3r^2) = 0 \end{aligned}$$

$$\frac{\partial F_1}{\partial x} = -c \frac{\frac{\partial x}{\partial x} r^3 - x \frac{\partial r}{\partial x}}{r^6}$$



EK



$$\begin{aligned}
 \text{volume} = & \left(\cancel{v_x} + \frac{\partial v_x}{\partial x} dx \right) dy - \cancel{v_x} dy \\
 & + \left(\cancel{v_y} + \frac{\partial v_y}{\partial y} dy \right) dx - \cancel{v_y} dx = \underbrace{\left(\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} \right) dx dy}_{\text{div } v}
 \end{aligned}$$

1.4 The curl operator (rotor, rotational)

1.4.1 Definition.

let be $\Omega \subset \mathbb{R}^n$ on open domain and

$$F: \Omega \rightarrow \mathbb{R}^n$$

$$x \mapsto F(x) = (F_1(x), F_2(x), \dots, F_n(x))$$

a vector field $F \in C^1(\Omega, \mathbb{R}^n)$

The curl operator denoted as $\text{curl } F$ (or $\nabla \times F$, or $\text{rot } F$)

• in 2D ($n=2$)

$$\text{curl } F(x, y) = \frac{\partial F_2}{\partial x}(x, y) - \frac{\partial F_1}{\partial y}(x, y) \in \mathbb{R}$$

• in 3D ($n=3$)

$$\text{curl } \vec{F}(x, y, z) = \begin{pmatrix} \frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z} \\ \frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x} \\ \frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \end{pmatrix} \in \mathbb{R}^3$$

$$\nabla \times \vec{F} = \begin{vmatrix} \vec{e}_x & \vec{e}_y & \vec{e}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_1 & F_2 & F_3 \end{vmatrix}$$





$$= \left(\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z} \right) \vec{e}_x + (\dots) \vec{e}_y + (\dots) \vec{e}_z$$

1.4.2 Examples

let us consider the gravitational force field with $\mathbf{p}_0 = (0, 0, 0)$

$$F: \mathbb{R}^3 \setminus \{(0, 0, 0)\} \rightarrow \mathbb{R}^3$$

$$(x, y, z) \mapsto F(x, y, z) = -\frac{c}{r^3} (x, y, z)$$

$$\frac{\partial F_2}{\partial x} = \frac{\partial}{\partial x} \left(-\frac{c}{r^3} y \right) = \frac{3cy}{r^4} \frac{x}{r}$$

$$\dots$$
$$\operatorname{curl} F = \frac{3c}{r^5} \begin{pmatrix} zy & -yz \\ xz & -zx \\ yx & -xy \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

